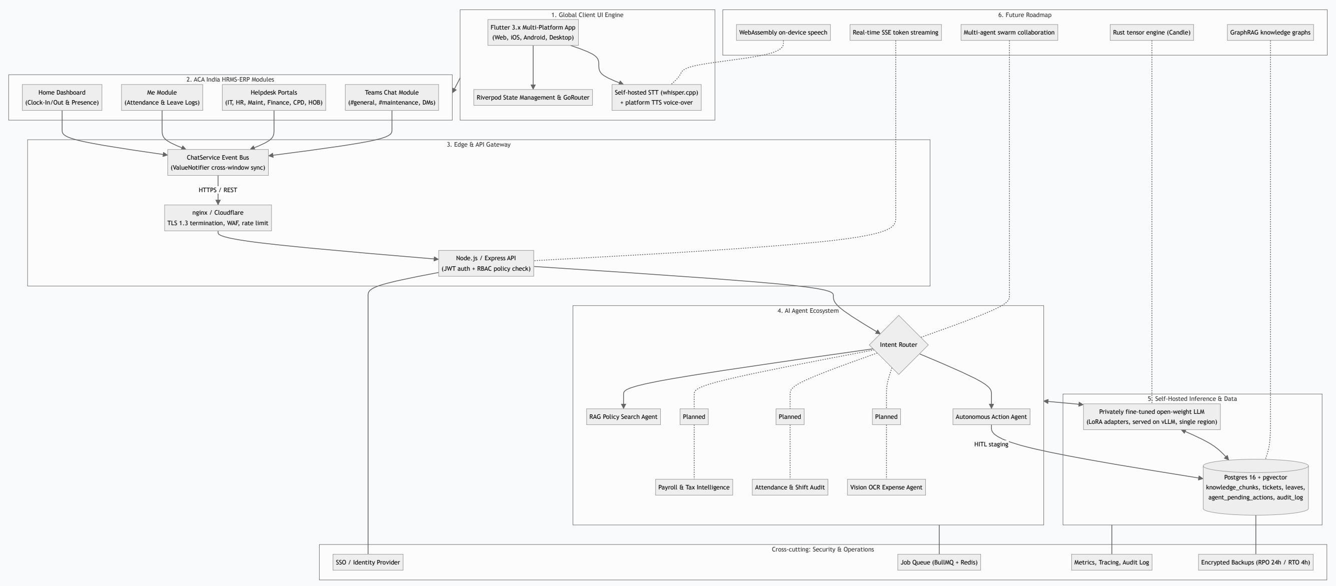
Asian Christian Academy of India

HRMS-ERP & Self-Hosted AI Engine Master Architecture Blueprint

Version 1.0.0 · Document Review: August 2026 · Status: Enterprise Blueprint

Self-Hosted Open Weights | Zero Cloud API Dependencies

1. Master System Architecture Chart



2. Custom AI Engine Development Stack & Infrastructure Engineering

The Asian Christian Academy of India AI Engine is developed as an in-house, self-hosted neural platform running privately fine-tuned open-weight models. The tech stack, codebase, and infrastructure components include:

1. Model Fine-Tuning Framework (Python, PyTorch & LoRA)

The core intelligence engine uses open-weight base models (e.g. Llama 3 / Gemma 2) fine-tuned on Asian Christian Academy of India HR policies, leave rules, and SOP documents using **PyTorch**, **HuggingFace Transformers**, and **PEFT / LoRA (Low-Rank Adaptation)**.

2. Dual-Tier Self-Hosted Inference Runtime (vLLM & llama.cpp)

Model serving is structured in two operational tiers: **vLLM (PagedAttention)** provides primary GPU serving for high concurrent campus traffic (achieving 30–150 ms time-to-first-token), with **llama.cpp (GGUF Quantization)** acting as a low-power CPU fallback server.

3. Private Self-Hosted STT & Native Platform TTS

To preserve complete data privacy, client voice input uses self-hosted **whisper.cpp / faster-whisper** speech-to-text models hosted on the local gateway, paired with native platform SpeechSynthesis voice-over engines on client devices.

4. Edge Security & TLS Termination (nginx & Express API Gateway)

All traffic from mobile, web, and campus devices is encrypted in transit using **TLS 1.3 HTTPS** terminated at an **nginx / Cloudflare edge node** (with WAF and rate limiting) before routing to the **Node.js / Express API service** with **JWT user authentication** and **RBAC policy enforcement**.

5. Unified Relational & Vector Storage (Postgres 16 + pgvector)

Relational ERP tables (employees, leaves, tickets, audit logs) and 1536-dimensional RAG document embeddings (`knowledge_chunks`) are stored in a unified **Postgres 16 + pgvector** database, guaranteeing transactional consistency and simplified single-node backup routines (RPO 24h / RTO 4h).

6. Cross-Cutting Operations (BullMQ, Redis & Security Services)

Long-running background tasks are managed via a **BullMQ + Redis** job queue. System health is monitored through open metrics and structured logging, backed by automated nightly AES-256 encrypted backups.

3. Asian Christian Academy AI Agent Ecosystem

Active Core Agents & Services

- **RAG Policy Search Agent (`ragService.js`)**: Grounded policy Q&A over indexed HR, IT, and leave SOP documents.
- **Autonomous Action Agent (`agentService.js`)**: Converts prompts into backend tool calls (`createTicket`, `applyLeave`, `sendMessageToUserOrChannel`).
- **Central Event Bus (`ChatService`)**: Cross-window synchronization service broadcasting live chat updates.
- **Voice I/O Interface (`VoiceHelper`)**: Client-side voice interaction wrapper managing mic audio and TTS playback.
- **HITL Safety Manager**: Staging service placing high-risk database mutations in `agent_pending_actions` for confirmation.

Planned Expansion Agents

- **Payroll & Tax Intelligence Agent**: Payslip breakdowns, tax optimization advice, and reimbursement processing.
- **Attendance & Shift Anomaly Agent**: Automatically detects clock-in anomalies and suggests regularizations.
- **Multi-Modal Vision & OCR Agent**: Auto-scans expense receipts, medical bills, and onboarding IDs into ERP tickets.
- **IT Diagnostic & Asset Specialist**: Monitors asset health, compares vendor pricing, and tracks PO approvals above ₹1 Lakh.
- **Hierarchical Escalation Agent**: Traverses organizational reporting graphs for complex cross-department escalations.

4. Programming Language Evaluation for Self-Hosted AI Infrastructure

Language	Evaluation & Suitability	Strengths for Self-Hosted AI	Target Architecture Role
Rust	5/5 — Recommended for Future Core Engine	Memory safety without Garbage Collection pauses, zero-cost abstractions, native C/CUDA interop, high-speed tensor libraries (Candle / Burn), WASM compilation.	**Evaluated for Future High-Throughput Native Inference Core**
Python	4/5 — Essential for Model Fine-Tuning	Unrivaled AI ecosystem (PyTorch, JAX, HuggingFace, vLLM). Standard language for dataset preparation and LoRA adapter fine-tuning.	**Dataset Preparation, Fine-Tuning (LoRA) & Training**
C++ / CUDA	4/5 — Low-Level GPU Kernels	Foundation of GGUF, llama.cpp, and TensorRT. Maximum low-level GPU hardware control.	**Low-Level CUDA GPU Kernel Execution**
Mojo	3/5 — Emerging AI Technology	Python superset compiling to native C speed with high-performance matrix math capabilities.	**Research Exploration**

Asian Christian Academy AI Architecture Strategy: A **Hybrid Microservice Stack**:
- **Python (PyTorch & vLLM)**: Model Fine-Tuning (LoRA) & High-Concurrency GPU Inference Serving.
- **C++ / Rust (llama.cpp / Candle)**: CPU Fallback Engine & Evaluated Future High-Performance Core.
- **Node.js / Express behind nginx**: Edge API Gateway, TLS 1.3 Termination, & Application Routing.
- **Dart (Flutter)**: Cross-Platform Client UI for Web, Mobile & Desktop.